Social Media Engagement Analytics Using Python

1. Project Overview

This project focuses on analyzing a social media engagement dataset using **Python, NumPy, Pandas, and data visualization libraries such as Matplotlib and Seaborn**. The goal is to perform **data cleaning, transformation, exploratory data analysis (EDA), and visualization** to uncover meaningful patterns and insights from the dataset.

Objective

- Data Cleaning
- Data Formatting
- Data Exploration
- Data Wrangling
- Statistical Analysis
- Data Visualization
- Insights Analysis

---

2. Data Sources

- **Source Description and Timeline:** Data government
- **Domain**:Entertainment

---

3. Problem Statement

Social media platforms generate massive volumes of engagement data—likes, comments, shares, impressions, watch time, and more. Analyzing such data helps companies understand user behavior, identify trends, and improve content performance.

In this assignment, you will work with a **social media dataset** containing social media engagement metrics. The task involves **data cleaning, transformation, NumPy/Pandas operations, exploratory data analysis, visualizations**, and generating insights.

---

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
User_ID	Integer	Unique number for each User
Age	Integer	Age of the user
Gender	String (Text)	Gender of the user
Country	String (Text)	Country of the user
Post_ID	Integer	Unique number for each post by the user

Post_type	String(Text)	Type of the post by the user
Post_category	String(Text)	Category of the post by the user
Likes	Integer	Likes for the post
Comments	Integer	Comments for the post
Shares	Integer	Shares for the post
Watch_time_sec	Integer	Watched time for the post in seconds
Impression_count	Integer	Impression count for the post
Posted_at	Date and time	Date and time of the post posted by the user
Follower_count	Integer	Followers count of the user
Is_verified	Boolean	Followers are verified for the user
Device_type	Integer	Type of device used for post
Sentiments	String(Text)	Type of sentiments for the user for the post
Hashtag	String(Text)	Hashtag used for the post by the user
Engagement_rate	Float	User engagement by the platform

---

5. Tools & Technologies

- **Python Google Colab Notebook:** Data cleaning, transformation, Visualization and analysing.

---

6. Data cleaning

- **Data Cleaning**

  Cleaning Missing Data

  Detect missing values (isnull(), isna())

  Handle using: dropna(), fillna(), median/mode, forward/backward-fill

  Duplicate Handling- Identify & remove duplicates

- ***Data Formatting***

  Fix incorrect data types

  Standardize categories (e.g., gender labels)

  Correct unrealistic values in likes, comments, shares

- ***Feature Cleaning***

Extract hashtag count

Clean sentiment labels

- **Data Exploration using Pandas**

  View dataset structure using head(), tail(), shape, and columns.

  Check data types and info with info() and dtypes.

  Generate summary statistics using describe().

  Analyze categorical distributions using value_counts(), unique(), and nunique().

  Create a correlation matrix for numeric fields.

  Use groupby() to summarize metrics (e.g., avg likes by post type, impressions by country).

- **Data Wrangling**

  Use merge, concat, or join if combining DataFrames.

  Create new fields such as *engagement_score, log-transformed metrics* (optional), and hashtag count.

  Perform groupby summaries by post_type, country, and sentiment.

- **Statistical Analysis**

  Mean, median, mode

  Standard deviation, variance

  Percentiles

  Skewness and kurtosis

- **Data Visualization**
- Matplotlib
  - Scatter: *likes vs impressions*
  - Line: *daily engagement trend*
  - Bar: *posts by category*
  - Pie: *gender distribution*
  - Histogram: *age*
  - Box: *engagement rate*
- Seaborn
  - Count plot: *post type*
  - Bar plot: *avg likes by category*
  - Violin: *followers vs sentiment*
  - Pair plot: *numeric features*
  - Heatmap: *correlation matrix*
  - Swarm plot: *engagement vs device*
- Plotly (Interactive)

- Interactive line chart/bar chart/bubble/scatter chart

---

9. Insights

Descriptive

- Engagement varies across posts based on likes, comments, shares, and impressions.
- Certain content types and verified accounts show higher engagement.
- Younger users and mobile users interact more with social media posts.

Diagnostic

- Engaging and visually appealing content attracts more interactions.
- Higher impressions increase visibility, leading to more engagement.
- Peak user activity times lead to higher interaction rates.

Predictive

- Posts shared during peak hours with engaging content will likely receive higher engagement.
- Content targeting younger audiences may continue to perform better.

Prescriptive

- Post engaging and visually rich content.
- Schedule posts during high user activity times.
- Use engagement insights to optimize future social media strategies.

---

10. Conclusions

The results show that factors such as **content type, posting time, user demographics, and sentiment** play a significant role in determining engagement levels. These insights can help organizations **optimize their content strategy, improve audience targeting, and maximize engagement on social media platforms**. Overall, this project demonstrates how data analytics can be used to convert raw social media data into **actionable insights for better decision-making**.